

Madis Altmann

Senior Python Full-Stack Engineer | FastAPI, Django, React, AWS, Microservices, AI/ML

CONTACTS

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EDUCATION

M.S., Computer Science

University of Tartu
Sep 2010 - Jun 2015
GPA: 4.6/5.0

COMPETENCIES

- Backend Engineering (Python)
- Full-Stack Development
- Distributed Systems & Microservices
- AI-Driven Product Development
- Real-Time & Event-Driven Systems
- Database & Data Engineering
- Asynchronous Processing & Performance Optimization
- Cloud & DevOps (AWS/GCP)
- Asynchronous Processing & Performance
- System Design & Architecture
- Agile Development & Collaboration

ARCHIEVEMENTS

- Reduced manual documentation by 70% by delivering real-time AI transcription and summarization.
- Scaled distributed systems to support 50+ concurrent real-time sessions with low latency and high reliability.
- Achieved millisecond-level search performance through advanced indexing and query optimization.
- Improved system responsiveness by implementing asynchronous, non-blocking processing pipelines.
- Increased system availability to 99.9%+ uptime with cloud-native, auto-scaling infrastructure.
- Boosted user engagement by 30% through AI-powered voice and conversational systems.
- Reduced operational workload by 75% via automation of core business workflows.
- Enabled real-time data processing using event-driven architecture and messaging systems.

SUMMARY

Senior Python & TypeScript Software Engineer with 10+ years of experience building scalable web applications and AI-powered platforms across healthcare, media, and enterprise communication domains, with strong expertise in backend engineering using Django, FastAPI, microservices, REST APIs, distributed systems, performance optimization, databases, and cloud-native architectures on AWS, GCP, and Azure, plus frontend development with React, Angular, and Vue focused on responsive UI, WCAG 2.2 accessibility, SEO best practices, and modern UX, alongside hands-on experience integrating GenAI solutions including Copilot, OpenAI, Claude, LangChain, RAG, and NLP, and delivering secure, production-ready systems with Docker, Kubernetes, CI/CD, JWT, OAuth2, RBAC, and Passwordless Magic Link in Agile environments.

WORK EXPERIENCE

Senior Full-Stack Software Engineer

Innowise | Paris, France | Apr 2023 - Mar 2026

SVENSA Voice - AI-powered conferencing platform

- Led backend engineering using **Python 3.11–3.12**, **Django 4.2+**, **AWS EC2**, **WebRTC (LiveKit)** to deliver real-time multilingual conferencing with AI transcription, reducing manual documentation by 70% and improving collaboration efficiency.
- Architected distributed systems with **FastAPI 0.100+**, **gRPC**, **RESTful APIs**, **DDD**, **OOP**, enabling low-latency communication pipelines and stable performance across 50+ concurrent sessions.
- Designed high-performance data and search layers using **RDBMS**, **PostgreSQL 14–16**, **SQLAlchemy**, **query optimization**, enabling millisecond-level transcript retrieval and analytics.
- Built asynchronous processing systems with **Redis 7.0**, **Celery 5.3+**, **pub/sub messaging**, **task orchestration**, supporting real-time AI inference and non-blocking workloads.
- Engineered cloud-native infrastructure using **Docker**, **Kubernetes (EKS 1.26+)**, **AWS ECR**, **CloudWatch**, **CI/CD pipelines**, **GitHub Actions**, ensuring scalability, observability, and resilient production environments.
- Integrated GenAI capabilities using **OpenAI APIs (GPT-4/4o)**, **LangChain 0.1+**, **RAG**, **Claude 3**, **Copilot**, **Speech-to-Text**, enabling real-time summarization and multilingual intelligence.
- Developed front-end features using **React 18**, **TypeScript**, **D3.js**, responsive UI, and **WCAG 2.2** standards while implementing advanced **SEO** with **Next.js SSR/SSG**, dynamic rendering, **React Helmet**, Core Web Vitals (**LCP**, **CLS**, **INP**, **FCP**, **TTFB**), schema markup, **XML** sitemap, robots.txt, canonical tags, **CDN** caching, and image optimization.

Senior Software Engineer

ELEKS | Tallinn, Estonia | Aug 2020 - Mar 2023

MemoryVoice - AI Voice Assistant (Healthcare / Digital Health)

- Developed secure backend services using **Python 3.9–3.10**, **FastAPI 0.70–0.95**, **AWS S3**, **IAM**, enabling voice-driven patient interaction with reliable storage and access control.
- Architected scalable microservices using **REST APIs**, **async processing**, **orchestration patterns**, **TDD**, **SOLID principles**, supporting conversational AI and contextual memory pipelines.
- Designed flexible data models using **MongoDB 4.4–6.0**, **DynamoDB**, **Elasticsearch 7.0**, **document schemas**, **advanced indexing**, enabling efficient patient history and contextual retrieval.
- Implemented low-latency pipelines with **Redis 6.0**, **Celery 5.2+**, **background job queues**, **caching strategies**, improving real-time voice response performance.
- Deployed secure infrastructure using **Docker**, **AWS EKS (1.21–1.24)**, **Lambda**, **encryption protocols**, ensuring privacy and regulatory compliance.
- Built AI systems using **NLP pipelines**, **Speech-to-Text**, **Text-to-Speech**, increasing patient engagement and routine adherence by 30%, and leveraging data processing tools such as **Pandas**, **NumPy**, **SciPy**, and **PySpark**.
- Developed dynamic applications using **Angular**, **RxJS**, reactive forms, and state management while applying technical SEO through semantic HTML5, metadata strategy, schema markup, sitemap optimization, canonical tags, mobile-first UX, and performance tuning, increasing usability and search visibility for healthcare portals.

SKILLS

Languages & Core Backend

Python, TypeScript, JavaScript, SQL, Object-Oriented Programming (OOP), SOLID Principles

Backend Frameworks & APIs

Django, FastAPI, REST APIs, gRPC, GraphQL, WebSockets, Microservice Architecture, Domain-Driven Design (DDD), CQRS, API Gateway Patterns, Clean Architecture, Middleware Orchestration, Serverless Architecture, WebRTC, LiveKit

Frontend Development

React, Next.js, Angular, Vue, D3.js, RxJS, Reactive Forms, State Management, Component-Based Architecture, SSR, SSG, SEO, UI/UX, Responsive UI, Tailwind CSS, Bootstrap, MUI, SCSS

AI / Machine Learning

LLM Integration (OpenAI GPT, Claude, Copilot), LangChain, RAG, NLP Pipelines, Intent Recognition, Speech-to-Text, Text-to-Speech, Pandas, NumPy, SciPy, PySpark, Behavioral Analytics, Ranking Models, Collaborative Filtering

Databases & Search

PostgreSQL, MySQL, MongoDB, DynamoDB, RDBMS, SQLAlchemy, Elasticsearch, Query Optimization, Indexing, Data Modeling, Document Schemas, Taxonomy Design

Cloud & DevOps

AWS (EC2, S3, Lambda, ECR, EKS, CloudWatch, IAM, Route53, CloudFront), Azure, GCP, Docker, Kubernetes (EKS), GitHub Actions, CI/CD Pipelines, High Availability, Multi-Region Deployment, Edge Delivery

Asynchronous & Messaging

Redis, Celery, Kafka, RabbitMQ, Pub/Sub, Background Processing, Task Orchestration, Async Processing, Event Triggers, Streaming Pipelines, Distributed Messaging, Caching Strategies

Monitoring & Observability

Grafana, CloudWatch, Observability, System Monitoring

Testing & Code Quality

Test-Driven Development (TDD), PyTest, unittest, Jest, React Testing Library, Integration Testing

Architecture & Performance

Distributed Systems, Event-Driven Architecture, High-Concurrency Systems, Performance Optimization, Real-Time Systems, Low-Latency Systems, Scalable System Design

Tools & Collaboration

Git, Agile (Scrum, Kanban), Cross-Functional Collaboration, Regulatory / Compliance-Aware Development, External System Integrations (PMS, CRM, POS)

Software Engineer

[Chudovo OÜ](#) | Narva, Estonia | Jan 2019 - Jul 2020

AI Voice Concierge - Hospitality & Travel Platform (SaaS / Automation)

- Delivered backend systems using **Python 3.7–3.8, Django 2.2–3.1, AWS Route53, cloud-native architecture**, automating guest interactions and reducing manual operations by 75%.
- Built integration layers with **FastAPI 0.45–0.63 (early adoption), API gateway patterns, PMS/CRM/POS integrations, middleware orchestration, clean architecture**, enabling seamless ecosystem connectivity.
- Optimized relational databases using **MySQL 5.7/8.0, PostgreSQL 11–12, query tuning, indexing strategies**, supporting high-throughput transactional workloads.
- Engineered event-driven systems with **Kafka 2.3–2.6, RabbitMQ 3.7–3.8, streaming pipelines, distributed messaging**, enabling real-time service orchestration and predictive workflows.
- Implemented serverless workflows using **AWS Lambda, event triggers, lightweight compute functions, asynchronous processing**, enabling scalable booking automation, notifications, and external system callbacks.
- Deployed resilient infrastructure using **AWS EKS (1.17–1.18), S3, high availability setups, global scaling**, achieving 99.95% uptime across distributed systems.
- Implemented AI automation using **classical NLP + intent recognition pipelines (pre-LLM era)**, delivering sub-400ms responses with high accuracy.
- Built multilingual user interfaces using **Vue.js**, component-driven UI, and adaptive layouts while delivering growth-focused SEO with localized landing pages, metadata automation, clean URLs, breadcrumbs, page speed tuning, and CTR optimization, improving guest experience, booking traffic, and regional rankings.

Full-Stack Engineer

[Ronas IT](#) | Tallinn, Estonia | Jul 2015 - Dec 2018

MediaSphere - AI Content Personalization Platform (Media / EdTech)

- Built scalable backend systems using **Python 3.5–3.7, Django 1.11–2.1, GCP, API-first architecture**, delivering unified content across web, mobile, and smart TV platforms.
- Designed modular services using **GraphQL, REST APIs, WebSockets**, service decoupling, **CQRS pattern**, enabling flexible and scalable content delivery.
- Developed data layers using **DynamoDB, MongoDB 3.4–4.0**, content modeling, taxonomy design, supporting large-scale personalized content systems.
- Implemented real-time pipelines with **Kafka 1.4–2.0, Redis pub/sub, event-driven processing**, enabling instant recommendation updates.
- Managed global infrastructure using **CloudFront, IAM**, multi-region deployment, edge delivery, ensuring high availability and low-latency access.
- Built AI personalization systems using **traditional ML (collaborative filtering, ranking models), behavioral analytics**, increasing engagement by 20%.
- Developed front-end platforms using **React 16, Next.js 7–9, SSR, SSG, performance optimization**, improving UX, SEO, and cross-device performance.

PROJECTS

SVENSA Voice - AI-powered conferencing platform:

<http://www.svensavoice.com>

Enterprise AI platform that combines video meetings, real-time translation, speech-to-text transcription, and automated meeting workflows into a single solution. It enables seamless multilingual communication and intelligent meeting management through scalable, cloud-based infrastructure.

AI Voice Concierge - Hospitality & Travel Platform (SaaS / Automation)

<https://www.concierge.bot>

AI-powered voice concierge platform that automates guest requests, bookings, and service workflows using conversational AI and predictive intelligence. It integrates with core hotel systems to deliver real-time, multilingual support and seamless service orchestration across hospitality operations.